

SHIREEN MEHER CHIRRAVURI

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SUMMARY

MS CS student at UMass Amherst (GPA: 4.0) with 4+ years building **scalable distributed systems** and cloud-native backend infrastructure. Engineered fault-tolerant, high-throughput microservices serving **8M+ merchants at 15K+ TPS**; built distributed systems on AWS with CI/CD and operational excellence. Skilled in Python, Java, Go, AWS, and scalable systems design.

SKILLS

Languages: Python, Java, Go, PHP, C, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: FastAPI, Flask, React, PyTorch, Hugging Face Transformers

Cloud & Infrastructure: AWS, RabbitMQ, Docker, Kubernetes, Jenkins, CI/CD, Git, Linux, REST/RESTful APIs, Object-Oriented Programming (OOP), Microservices, Distributed Computing

Databases & Search: PostgreSQL, Redis, MongoDB, Neo4j, Elasticsearch, NoSQL

ML & AI: Generative AI (Gen AI), TRL, GRPO, vLLM, RAG, LLM Training, Synthetic Data Generation, Weights & Biases, GPU Cluster Training

Relevant Coursework: Data Structures & Algorithms, Network & Computer Security, Database Design & Implementation, Distributed Operating Systems

EXPERIENCE

AI Research Intern | *Meta*

Jan 2026 – Jun 2026

- Engineered a **large-scale GPU cluster training pipeline** in Python; automated SLURM job submission, built data generation and reward extraction components with logging, monitoring, and experiment tracking across curriculum-scheduled runs and OOD evaluation sets, supporting RL fine-tuning research targeting **LLM reasoning and hallucination reduction**.

Software Engineer II | *Venwiz*

Mar 2023 – Jan 2025

- Architected cloud-native vendor discovery pipeline (Lambda/SNS/SQS, **RabbitMQ** for async inter-service messaging) decoupling SERP scraping from Elasticsearch indexing, preventing scrape failures and traffic spikes from degrading search availability.
- Designed **three-mode hybrid retrieval** for industrial procurement queries: semantic vector search, Lucene kNN/keyword scoring with weighted field relevance, and Boolean matching over must/should/must_not clauses.
- Owned Neo4j graph infrastructure for vendor-category relationships (APOC traversals, composite indexing, schema evolution); benchmarked Neo4j+Elasticsearch vs Elasticsearch-only and consolidated where recall was comparable.
- Developed a GPT-based metadata enrichment pipeline and Flask web app with Retool UI for self-serve taxonomy management; **cut search-ops turnaround from 3 days to under 30 minutes**.

Software Engineer | *Razorpay*

Jul 2021 – Feb 2023

- Reduced P99 latency 45%** (820ms to 450ms) on payments auth path serving 8M+ merchants and INR 5T+ GMV via goroutine concurrency, DB access redesign, and replica-backed reads at 15K+ TPS peak.
- Shipped Rule-Based Method Enablement: event-driven eligibility engine processing **500K+ merchant events/day**, auto-configuring payment methods via business rules, cutting activation from days to minutes for 95% of cases.
- Led payment-terminal service decomposition migrating 12+ read/write paths to service ownership; drove unit and integration testing; resolved 341 cross-module test failures preserving transaction correctness.
- Increased **merchant activation success from 91% to 99.3%** by offloading reads to replica paths, cutting primary DB contention 60% and enabling horizontal scale-out without reliability regressions.

PROJECTS

Grounded Entity Search

Jan 2026 – May 2026

github.com/ShireenMeher/grounded_entity_search

- Built a full-stack, source-grounded entity search web app (**Python/FastAPI** backend, **React** frontend, **RESTful APIs**) turning natural-language queries into structured entity tables via an 8-stage LLM pipeline with hallucination detection (**97% evidence verification**); parallelization cut latency from 140s to 30–40s.

Distributed Workflow Orchestrator

May 2026 – Jul 2026

github.com/ShireenMeher/distributed-workflow-orchestrator

- Built a DAG workflow orchestration engine in **Go** (API/Scheduler/Worker binaries); used **Postgres as the queue** via SELECT FOR UPDATE SKIP LOCKED, eliminating an external broker; implemented **worker lease and heartbeat** for crash detection and a stateless scheduler that auto-resumes on restart.

Database Storage Engine

Jan 2026 – Apr 2026

github.com/ShireenMeher/database_engine

- Built a relational storage engine in **Java**: **buffer pool with LRU eviction**, B+ tree index with sibling pointers, and volcano-model query executor supporting joins and materialized projections.

EDUCATION

University of Massachusetts Amherst

Jan 2025 – Dec 2026 (Expected)

Master of Science in Computer Science, GPA: 4.0/4.0

Indian Institute of Technology Bhubaneswar

2017 – 2021

Bachelor of Technology